Report R12 — the sectors in the matrix: block-diagonal (WCC) and Frobenius normal (SCC) form

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

Every fragmentation statement in R2 and R9 is a number: a sector count, a growth base, a sub-leading exponent. Those numbers describe a property of a matrix — that the one-step transition matrix is block-diagonal in a basis nobody had drawn — and this report draws it. Two spy plots of the *same* matrix for rule 156 (`IIVI`) at `obc0`: once in the computational basis, once with the basis permuted so that each weak component is contiguous. The permuted picture is a staircase of 144 disjoint blocks at $N = 10$, the largest holding 20 of 1024 states, and **not one nonzero lies outside a block** — checked, at every N from 8 to 12, rather than asserted. Three quantitative readings come out of the same computation: the block count is exactly the Fibonacci number F_{N+2}, which is where R2's golden ratio comes from; the matrix thins from 1.81% to 0.24% dense between $N = 8$ and $N = 12$ while the *interior* of the blocks stays roughly two-thirds full, so the concentration factor grows from 41× to 259×; and the computational-basis panel is emphatically not structureless — it is strongly banded and self-similar — it simply does not show *sector* structure.

§4 then reorders the same matrix into its *Frobenius normal form* — strongly connected components in reverse topological order, *nested inside* the sector blocks, so the picture is block-diagonal at the sector level and block-upper-triangular within each sector — for rules 156, 157, 109, 150 and 158, chosen to span both axes: exponential sector counts (φ, ρ, ψ) and polynomial ones, unitary and dissipative. Nothing lies below the block diagonal in any of them. For the two unitary rules the triangular form *is* the diagonal form (R9's A2, drawn); for the three dissipative ones the strictly upper part is most of the matrix — 95% of it for rule 158. The pair 150/158 is the sharpest case: they share the *identical sector partition* at every N tested, so no sector observable can distinguish them, while their Frobenius forms are 6 strong components against 936. `obc0` only; `pbc` is held.

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1 What is plotted, and why

Definition 1 (the matrix)**.** For a rule, a length N and a boundary condition, let

$$M_{yx} = 1 \quad \Longleftrightarrow \quad \text{the one-step map carries } |x\rangle \text{ to } |y\rangle \text{ with nonzero amplitude.}$$

M is the support of the circuit unitary U, equivalently of the doubly stochastic $|U|^2$; the two have the same pattern, which is all a spy plot uses. Amplitudes are deliberately discarded — the point of the figure is that the *support alone* already carries the fragmentation.

Definition 2 (the permutation)**.** Let the weak components of the transition graph be $S_1, \ldots, S_K$, ordered by size, largest first, with states ascending inside each. Let P be the permutation that lists the basis in that order. The second panel plots PMP^{-1}.

Why rule 156**.** It is unitary (`IIVI`: two identities, a Hadamard, an identity), so $|U|^2$ is doubly stochastic, there are no transient states, and the weak, strong and forward-closure partitions all coincide — R9's assertion A2. The blocks are therefore unambiguous, and there is no question of whether we are drawing sectors or attractors: for this rule they are the same objects. It is also the rule R2 solves by room-packing, so its numbers are known independently of anything here.
Why it is worth a figure. "The dynamics block-diagonalises" is the central claim of the whole sector programme, and until now it has appeared only as counts and exponents. A permutation is not a theorem, but it makes the claim visible in one glance, and it makes the *failure* of the computational basis to reveal it equally visible.

2 The two panels

Three things to read off Figure 1.

- **The sorted panel is exactly block-diagonal.** Every one of the 6930 nonzeros lies inside one of the 144 shaded squares. Nothing is off-diagonal, at any N tested (§3).
- **The blocks are small and there are many of them.** The largest holds 20 of 1024 states, so the staircase is a thin diagonal thread rather than a few fat blocks. That is $D_{\max}/2^N \approx 0.02$ and falling — the quantitative form of "strong fragmentation", and the reason R2's $b = 4^{1/5}$ sits so far below the ergodic $b = 2$.
- **The left panel is *not* structureless.** It is strongly banded, with a self-similar block-of-blocks pattern at several scales. That is real and has an explanation — the integer order groups states by their high bits, and the rule is local, so states that agree on most sites sit near one another — but the bands run *across* sectors. Ordering by integer value reveals locality, not conservation. The contrast between the panels is not structure-versus-noise; it is the wrong structure versus the right one.

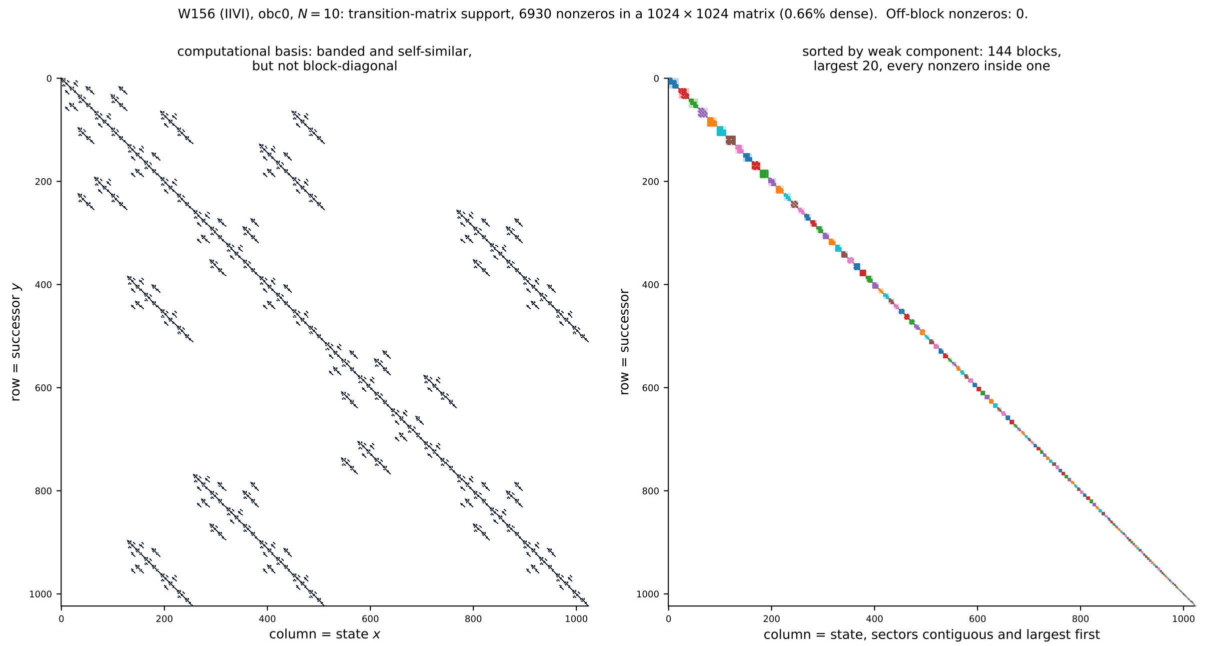

Figure 1: **S1, the same matrix in two bases** (rule 156, `obc0`, $N = 10$; 1024×1024, 6930 nonzeros, 0.66% dense). **Left:** computational basis, i.e. states ordered by their integer value. **Right:** basis permuted so each weak component is contiguous, largest first; the shaded squares are the components and the dots are coloured by which component they belong to. The two panels contain *identical* data. The right panel's caption number — off-block nonzeros = 0 — is computed, not assumed.

2.1 A closer look

At $N = 10$ the largest block is 20 states wide, which is about four pixels in Figure 1, so the interior of a block cannot be seen there. Figure 2 takes the first 140 rows and columns.
The blocks are *not* full: within a block the fill is about two-thirds (§3), and the interior pattern is itself structured rather than random. So a sector is not a featureless invariant subspace — it has its own sparse combinatorics, which is what the room-packing derivation of R2 is about. A smaller system makes the same point without a zoom:

3 Numbers

N	dim	nnz	density	blocks	$D_{\max}$	off-block
8	256	1189	1.81%	55	12	0
9	512	2870	1.09%	89	16	0
10	1024	6930	0.66%	144	20	0
11	2048	16731	0.40%	233	27	0
12	4096	40391	0.24%	377	36	0

Off-block nonzeros are zero at every size. That column is the whole content of the sorted panel, and it is a computation (`spy.check_block_diagonal`), not a restatement of the definition: the blocks are found by weak connectivity on the transition graph, and the check then asks independently whether any matrix entry crosses between them. A nonzero there would mean the partition and the matrix disagree.
The block count is Fibonacci. For $N = 6 \dots 12$ the number of sectors is

$$21,\ 34,\ 55,\ 89,\ 144,\ 233,\ 377\ =\ F_{N+2},$$

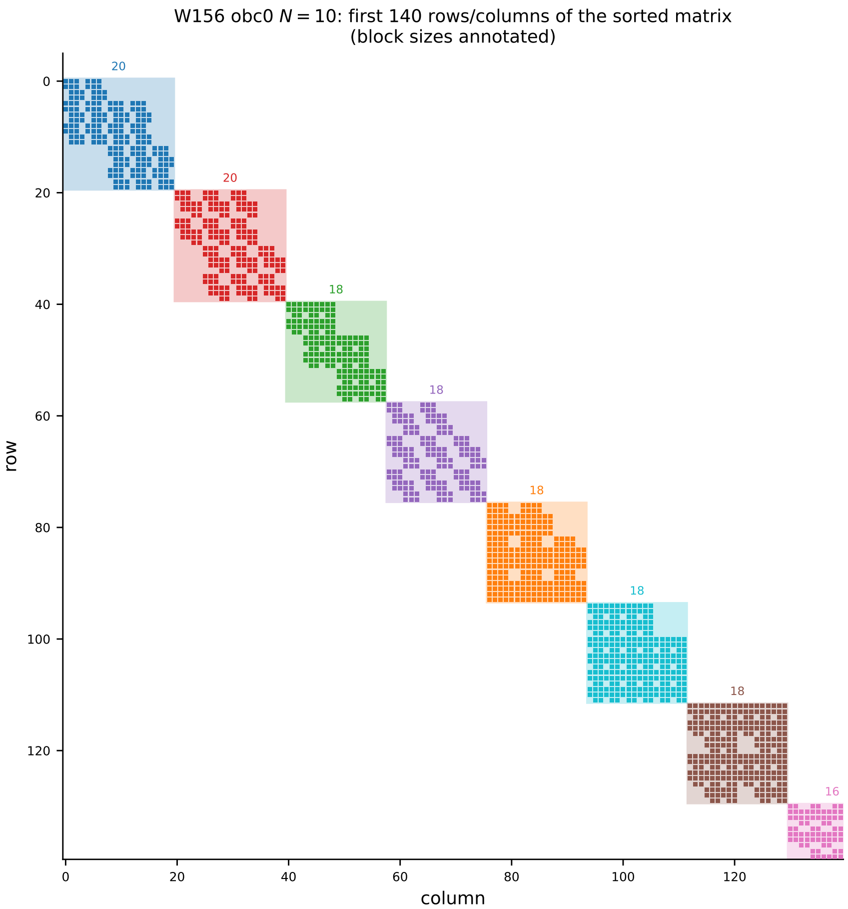

Figure 2: **S2, the top-left corner of the sorted matrix** (rule 156, `obc0`, $N = 10$), block sizes annotated. The largest sectors are 20, 20, 18, 18, 18, 18, 18, 16, ... states. Each block has its own internal sparsity pattern, itself visibly self-similar.

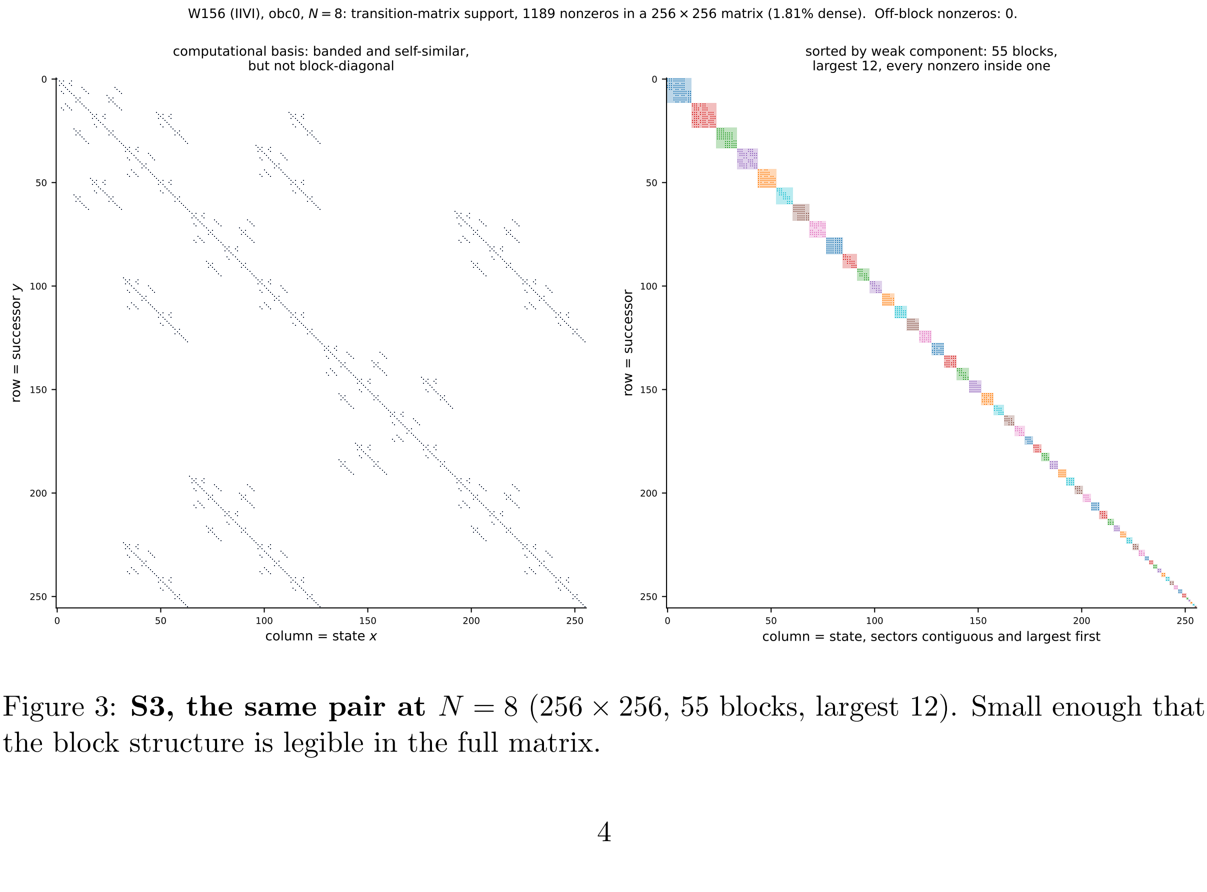

Figure 3: **S3, the same pair at** $N = 8$ (256×256, 55 blocks, largest 12). Small enough that the block structure is legible in the full matrix.

exactly, with $F_1 = F_2 = 1$. Hence the growth base $\varphi = 1.618034$ that R2 reports for this rule's sector count at `obc0`. (Under `pbc` the same rule gives $L_N + 1$ with L the Lucas numbers — a different sequence with the same base, which is why the boundary condition has to be stated with the constant.)

The nonzero count is a third constant, exactly. Counting the edges rather than the sectors gives, for $N = 6 \dots 14$,

$$204,\ 493,\ 1189,\ 2870,\ 6930,\ 16731,\ 40391,\ 97512,\ 235416,$$

which satisfies the integer recurrence

$$a_N \ = \ 2a_{N-1} + 2a_{N-3} + a_{N-4},$$

whose characteristic polynomial factors as

$$x^4 - 2x^3 - 2x - 1 \ = \ (x^2 - 2x - 1)(x^2 + 1).$$

The dominant root is the *silver ratio* $1 + \sqrt{2} = 2.414214$, and the $\pm i$ from the second factor is a pure period-4 modulation — which is exactly the $-1, -1, +1, +1$ pattern by which a_N misses $2a_{N-1} + a_{N-2}$. So this one rule carries three different algebraic constants at once, one for each question asked of the same matrix: φ for how many blocks there are, $4^{1/5}$ for how big the largest gets (R2), and $1 + \sqrt{2}$ for how many nonzeros the matrix has. None of the three is a fit; all are exact recurrences or derivations.

Thinning globally, not locally. As N grows the matrix becomes rapidly sparser in absolute terms, but the blocks do not hollow out:

N	density of M	fill *inside* the blocks	ratio
8	1.81%	74.1%	41×
10	0.66%	68.0%	103×
12	0.24%	62.3%	259×

The middle column is the nonzero count divided by $\sum_k |S_k|^2$, the total area available inside the diagonal blocks. It drifts down slowly while the global density falls by a factor of 7.5, so essentially all of the sparsity of M is the block structure and almost none of it is sparsity within a sector. That is the precise sense in which the permutation is the right change of basis: it accounts for the emptiness.

4 The Frobenius normal form

The WCC permutation makes M block-*diagonal*. The finer SCC partition makes it block-*triangular*, which is the Frobenius (irreducible) normal form of a non-negative matrix. The two coincide exactly when there are no transient states, and that is the same condition as unitarity — so this section is where the unitary/dissipative distinction becomes a picture.

4.1 The normal form, and when it is triangular

Definition 3 (reducible, irreducible)**.** A non-negative $n \times n$ matrix M is *reducible* if some permutation P brings it to $PMP^{-1} = \left(\begin{smallmatrix} A & B \\ 0 & C \end{smallmatrix}\right)$ with A and C square and non-empty, and *irreducible* otherwise. Frobenius' criterion: M is irreducible iff its directed graph — an edge $x \to y$ whenever $M_{yx} \neq 0$ — is *strongly connected.*

Proposition 1 (Frobenius normal form)**.** *Every non-negative M admits a permutation P with*

$$PMP^{-1} = \begin{pmatrix} M_{11} & M_{12} & \cdots & M_{1k} \\ 0 & M_{22} & \cdots & M_{2k} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & M_{kk} \end{pmatrix},$$

where each diagonal block M_{ii} is irreducible or is a 1×1 zero. The diagonal blocks are exactly the strongly connected components of the graph and are unique up to relabelling; the freedom in P is the choice of a topological order of the condensation, plus the order of states inside each block.

Two consequences decide what the figures can show.

When there is nothing to triangularise. If the graph is strongly connected there is a single component, $k = 1$, and the normal form is M itself. No permutation can split an irreducible matrix into two blocks with a zero corner — that is precisely what irreducible means. The same holds block by block: an individual M_{ii} is irreducible, so it has no internal block-triangular structure either.

Note what this does *not* say. An irreducible block still has plenty of nonzeros above its main diagonal in the elementwise sense — rule 150's 462×462 block is one — and the figures show them. Triangularity here is always a statement about *blocks*: whether the states can be split into two groups with no way back from the second to the first. Inside one irreducible block there is always a way back, by definition, so the elementwise pattern above the diagonal carries no drainage information at all. This is why Figures 4 and 7 look untriangular: for a unitary rule each sector *is* one component.

When the strictly upper part is nonzero. Not merely when $k \geq 2$. Two components with no edge between them give a block-*diagonal* form, not a triangular one. The strictly upper part is nonzero precisely when the condensation has at least one edge, i.e. precisely when some state is *transient* — it can leave its component and never return. So

$$\text{genuinely triangular} \iff \text{transient states exist} \iff \text{the rule is dissipative here,}$$

the last step because a unitary rule has $|U|^2$ doubly stochastic, which forbids a state from being unable to return. At $N = 10$ rules 156 and 150 have *zero* condensation edges and are block-diagonal; the other six have many.

Trivial blocks, and a warning about the diagonal. A singleton component with no self-loop gives a 1×1 *zero* block. These dominate the dissipative rules — 669 of 758 components for rule 157, 880 of 1024 for rule 100 — and they are why those pictures have so little on the diagonal and so much above it. Conversely a nonzero *on* the diagonal means only $x \to x$, not recurrence: rule 100 has 144 states with a self-loop but only 60 terminal components, because the other 84 can also step elsewhere and never come back. A self-loop makes a state a fixed point of one branch, not an attractor.

4.2 The ordering used here

Definition 4 (Frobenius order)**.** Take the strongly connected components of the transition graph and order them by a *reverse* topological order of the condensation: sinks — the terminal SCCs, i.e. the monitored attractors — before their predecessors. An edge then runs from a higher index to a lower one, so every nonzero sits on or *above* the block diagonal and weight drains up and to the left.

The two forms nest, and the figure is drawn so that they do. A reverse-topological order is not unique, and the choice matters here. Every condensation edge joins two SCCs of the *same* weak component — an edge is itself a witness of weak connectivity — so the condensation is a disjoint union of DAGs, one per sector, with no edges between them at all. Consequently any order that is reverse-topological *within each sector* is reverse-topological globally, whatever order the sectors are placed in. Taking that freedom to keep each sector contiguous, laid out largest first exactly as in the WCC panel, gives the **nested** form:

block-diagonal at the sector level,   block-upper-triangular inside each sector.

The third panel therefore carries the second panel's block structure — the sector outlines are drawn on it — with each block resolved into its internal triangular refinement. *Nothing ever crosses between sectors*; that is a theorem, not an artefact of the layout.

A global sweep that interleaves SCCs from different sectors by size is an equally valid Frobenius form and equally triangular, but it scatters each sector's states across the index range, so intra-sector drainage lands far from the diagonal and reads as though weight were crossing between sectors — which cannot happen. An earlier version of these figures did exactly that and was misleading; `spy.scc_blocks` now nests by default (`nest_in_wcc=True`) and the option to sweep globally survives only for §4.5's corner figures, which want every sink at the front.
The ordering is produced by a Kahn sweep on the condensation (`spy.scc_blocks`) rather than by trusting Tarjan's internal numbering. Tarjan does happen to finalise a component before its predecessors, but that is an implementation detail of the traversal, and the triangularity of the figure should not depend on it; `spy.check_frobenius` then verifies the result independently by counting the nonzeros below the block diagonal.

rule	tuple	n_{wcc}	n_{scc}	terminal	$\lvert\mathrm{Rec}\rvert$	in blocks	upper	below
156	`IIVI`	144	144	144	1024	6930	0	0
157	`EIVI`	16	758	16	67	1733	5414	0
109	`EIIV`	54	723	54	355	5131	6042	0
150	`IVVI`	6	6	6	1024	59049	0	0
158	`IEVI`	6	936	6	31	374	7545	0

(`obc0`, $N = 10$. "in blocks" counts nonzeros inside an SCC, "upper" the strictly upper part — the drainage — and "below" is the number that would falsify the triangular form.)
The five rules were chosen to span the two axes that matter. Sector count: exponential for 156 (φ), 157 (ρ) and 109 (ψ); polynomial (linear) for 150 and 158. Dynamics: unitary for 156 and 150, dissipative for 157, 109 and 158.

4.3 What the three panels show

- **Nothing lies below the block diagonal, for any of the five**, and **nothing crosses between sectors either** — both computed, not assumed. The sector outlines drawn on the third panel are the same blocks as the second panel's, so the two can be read against each other directly: the Frobenius form refines the block-diagonal form and never violates it.

- **For a unitary rule the triangular form *is* the diagonal form.** Rules 156 and 150 have $n_{\mathrm{scc}} = n_{\mathrm{wcc}}$, every SCC terminal, all 1024 states recurrent and a drainage of exactly zero. That is R9's assertion A2 as a picture: with $|U|^2$ doubly stochastic there is nowhere for weight to drain to, so weak and strong components cannot differ.

  So their blocks are not triangular, and should not be. Each sector is a *single* strong component — irreducible — and the Frobenius normal form of an irreducible matrix is the matrix itself, a 1×1 block structure with no strictly upper part to show. Triangularity is visible only where there is drainage to make it, i.e. only for the dissipative rules. In Figures 4 and 7 the whole of every sector is gold, which is the same statement in the other notation.

- **For a dissipative rule they are nothing alike.** Rule 158 has 6 weak components and 936 strong ones; 157 has 16 and 758; 109 has 54 and 723. The recurrent set is a thin gold corner — 31 states of 1024 for 158, 67 for 157 — and the bulk of the matrix is the grey drainage cloud filling the upper triangle. For 158, 7545 of 7919 nonzeros (95%) are strictly upper: almost the entire matrix is transient flow, and only 5% of it is the dynamics *inside* an attractor.

- **One attractor per sector, in all five.** $n_{\text{terminal}} = n_{\text{wcc}}$ exactly for every rule here, which is the generic case of R9 §7.3 — none of these five is among the twelve rules whose sectors carry more than one attractor. So the coarsest partition (WCC) and the set of sinks are in bijection, and everything between them is transient scaffolding.

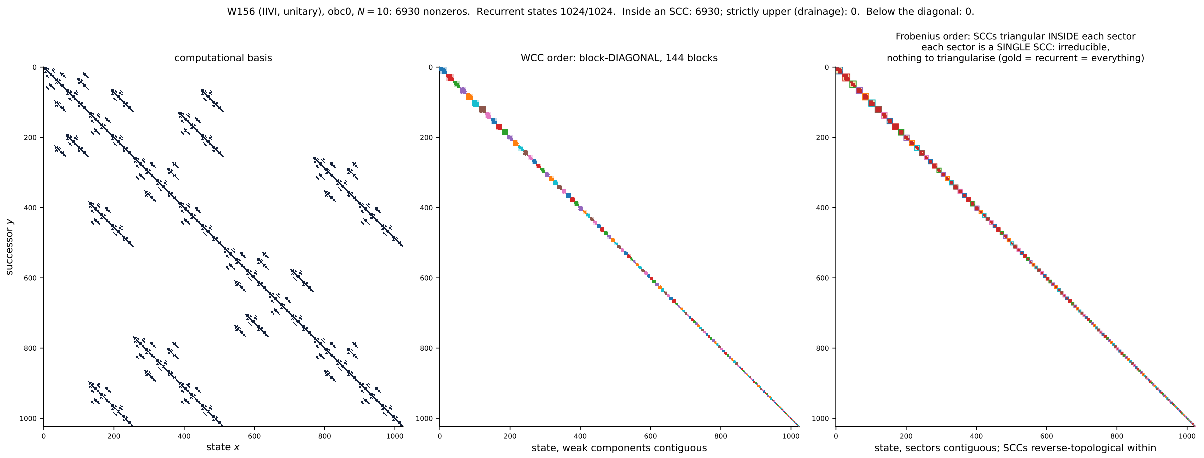

Figure 4: **S4, rule** 156 (`IIVI`**, unitary,** φ**).** Left: computational basis. Middle: WCC order, block-diagonal. Right: Frobenius order. All 144 SCCs are terminal, so the third panel is block-diagonal too and the drainage is empty — for a unitary rule the two normal forms coincide.

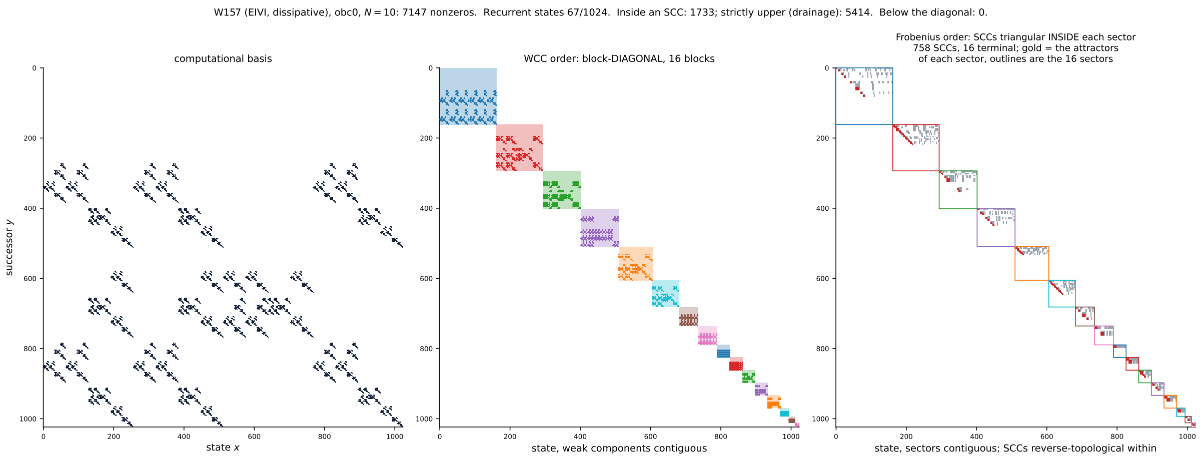

Figure 5: **S5, rule** 157 (`EIVI`**, dissipative,** ρ**).** 16 weak components but 758 strong ones. Gold marks the recurrent corner (67 states); red are the entries inside an SCC, grey the drainage.

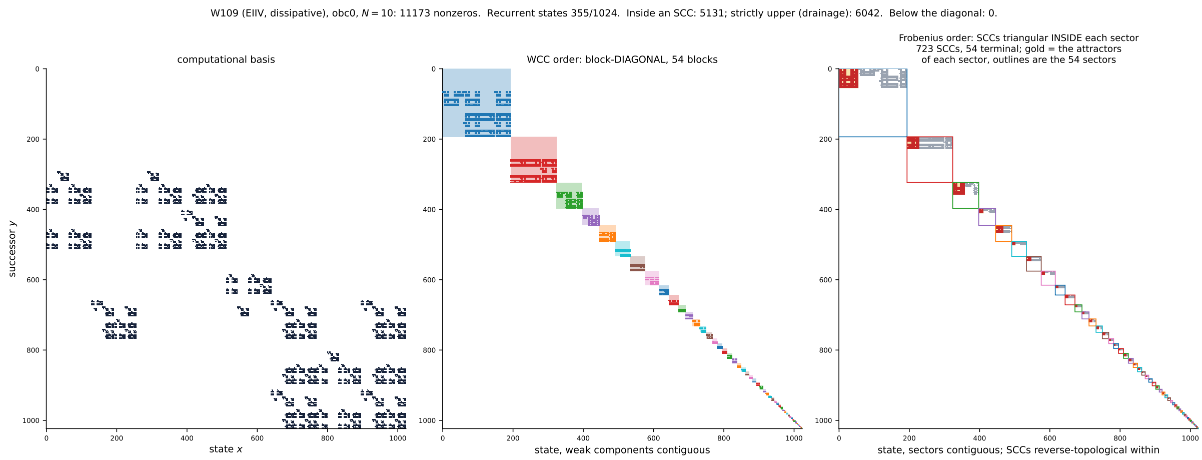

Figure 6: **S6, rule** 109 **(**`EIIV`**, dissipative,** ψ**).** The least contracting of the three dissipative rules: 355 of 1024 states recurrent and a largest SCC of 55, so the gold corner is a third of the matrix and the diagonal blocks are visible in their own right.

4.4 Rules 150 and 158: the same sectors, different matrices

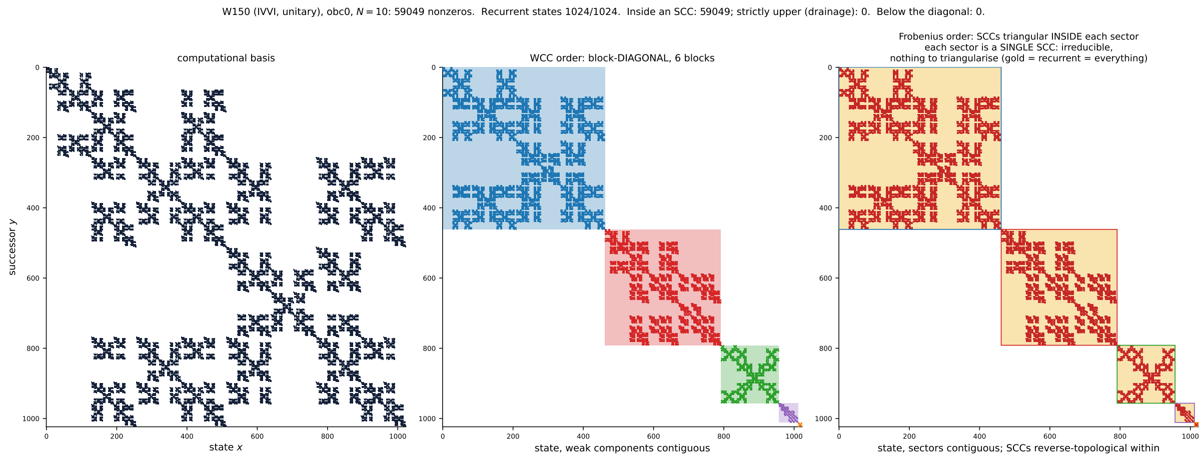

Figure 7: **S7, rule** 150 **(**`IVVI`**, unitary, linear sector count).** Six sectors of sizes $462, 330, 165, 55, 11, 1$; every one is a single SCC.

This pair is the sharpest statement in the report. Rules 150 and 158 do not merely have the same sector *count*: at $N = 8$, 10 and 12 they have the *identical sector partition*, the same sets of states, with sizes $462, 330, 165, 55, 11, 1$ at $N = 10$. Their middle panels are the same picture. Yet one is unitary with all 1024 states recurrent in 6 strong components, and the other retains 31 recurrent states in 6 terminal components out of 936. Nothing on the sector axis — not the count, not the sizes, not the partition itself — can tell them apart; the Frobenius form separates them completely. That is the concrete form of R9's finding that the sector axis is uninformative for the V+reset family, and the reason this project carries two maps rather than one.

4.5 Multi-attractor rules: one sector, several sinks

All five rules above have $n_{\text{terminal}} = n_{\text{wcc}}$, one attractor per sector, which is the generic case: R9 §7.3 finds it holds for 237 of 249 rules. The remaining twelve are the ones where the Frobenius form is strictly finer than the block-diagonal one in a way no sector count can report — a single

Figure 8: **S8, rule 158 (`IEVI`, dissipative, linear sector count).** The *same six sectors as rule* 150 — middle panel — shattered into 936 strong components of which 6 are terminal, holding 31 states between them.

weak component containing several terminal SCCs. All three sector-count classes occur among them, so here is one of each.

rule	tuple	n_{wcc} class	n_{wcc}	n_{scc}	terminal	|Rec|	max att/sector	upper
203	`VEII`	constant	1	969	28	34	28	98%
100	`IDIV`	polynomial	13	1024	60	60	19	97%
29	`EIVD`	exponential	7	949	16	67	5	93%

(`obc0`, $N = 10$; "upper" is the drainage as a percentage of all nonzeros.) At full scale these rules' attractors are a few dozen states inside a 1024-wide matrix, so the grouping cannot be seen in a three-panel figure. Figures 9–11 therefore show only the recurrent corner: the $|\text{Rec}| \times |\text{Rec}|$ top-left block, with each terminal SCC outlined and *coloured by the weak component that contains it.* Blocks sharing a colour inside a dashed bracket are several attractors inside one sector.

Reading the gold. In the three-panel figures the recurrent states are shaded gold *per sector*, not as one corner of the whole matrix. The nested ordering puts each sector's sinks at the start of *its own* range, so the attractors of a sector are a gold square in that sector's top-left corner and the transient states of the same sector fill the rest of its block. For a unitary rule every sector is entirely gold; for rule 158 the gold is 31 states spread over six sectors and is correspondingly hard to see, which is itself the finding.

The ordering inside the recurrent corner needs one remark. Terminal SCCs are sinks — no outgoing condensation edges — so *any* order among them still places every component before its predecessors, and the form stays triangular. These three figures use that freedom to group each sector's attractors together, which is what lets the brackets be drawn without overlapping; `spy.check_frobenius` confirms the below-diagonal count is still zero with the regrouping applied.

Rules 29 and 157: the same attractors, different sectors

Rule 157 already appears above (Figure 5) with 16 attractors in 16 sectors, one apiece. Rule 29 has 16 attractors in 7 sectors. They are *the same* 16 *attractors* — the identical sets of states, holding the identical 67 recurrent states — verified at $N = 9$, 10 and 11 (12 attractors/41 states, 16/67, 21/99). The two rules differ by a single symbol, `EIVD` against `EIVI`, and what the reset changes is not which attractors exist but how many sectors enclose them.

Set beside §4.4, this gives a matched pair of blind spots:

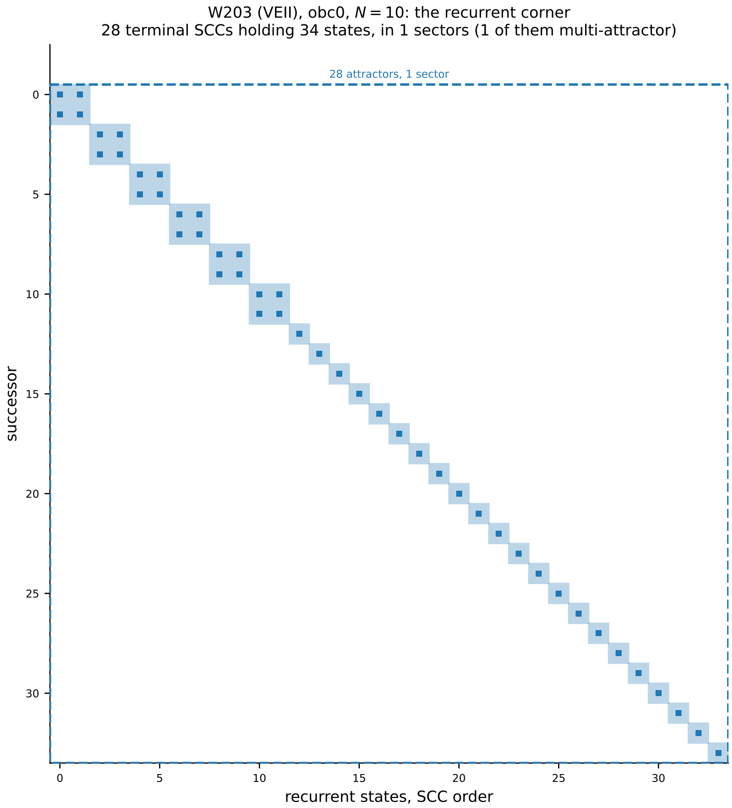

Figure 9: **S9, constant sector count: rule** 203 (`VEII`)**.** $n_{\mathrm{wcc}} = 1$ at *every* N — the dynamics connects the whole basis — and that one sector holds 28 attractors: six 2-cycles and 22 fixed points, 34 states in all. The sector observable returns 1 here whatever N is; the attractor count grows. This is the maximal degeneracy of the sector axis.

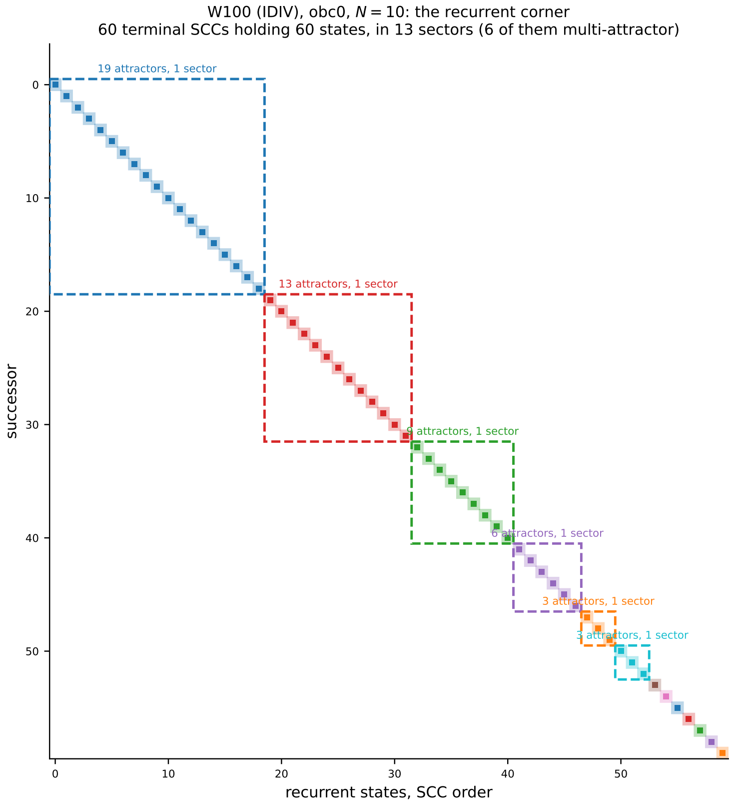

Figure 10: **S10, polynomial sector count: rule** 100 **(**`IDIV`**).** $n_{\mathrm{wcc}} = N + 3$ exactly $(9, 10, \ldots, 15$ for $N = 6 \ldots 12)$, so 13 sectors at $N = 10$ — holding 60 attractors between them, one sector taking 19. Every strong component is a singleton ($n_{\mathrm{scc}} = 1024 = \dim$), so the graph has no cycles at all and every attractor is a fixed point.

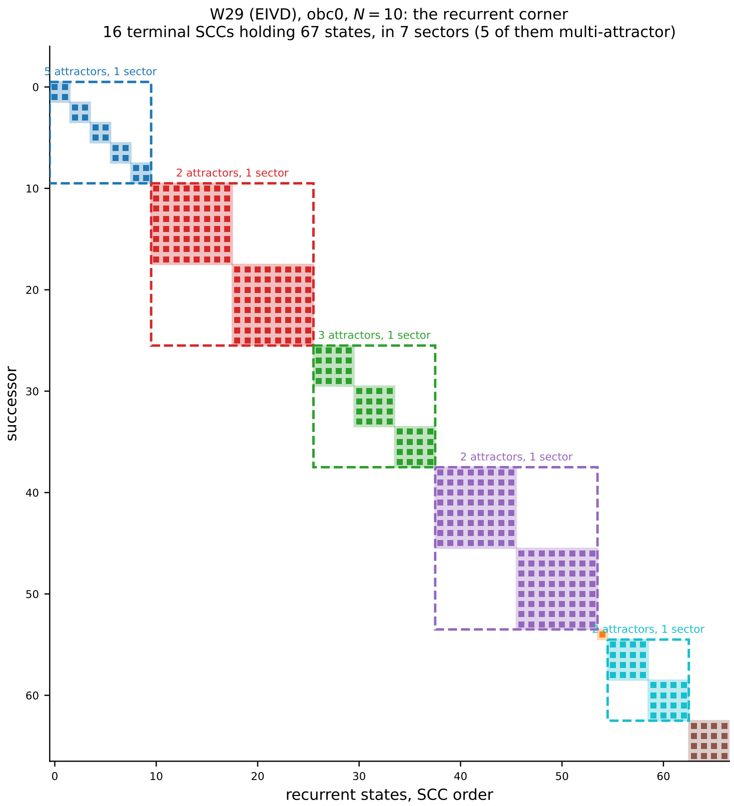

Figure 11: **S11, exponential sector count: rule** 29 **(`EIVD`).** 16 attractors in 7 sectors, five of them multi-attractor, with brackets of 5, 3, 2, 2 and 2.

pair	agree on	differ in
150 / 158	the sector partition, exactly	the attractors (6 vs 936 SCCs)
29 / 157	the attractor set, exactly	the sectors (7 vs 16 WCCs)

Neither observable dominates the other. A measurement that sees only sectors cannot separate 150 from 158; one that sees only attractors cannot separate 29 from 157. Both maps are needed, which is the conclusion R9 reaches by counting and this report reaches by looking.

5 How to read a block

The figures have three kinds of region and it is worth saying exactly what each one means, because one natural expectation about them is wrong.

A column is a state. Column x carries a nonzero in row y for every y the one-step map can produce from x. Its weight — the number of nonzeros — is therefore the *branching number* of that state. A site whose gate is a Hadamard and whose neighbours trigger it splits $|x\rangle$ into two successors, so k simultaneously firing Hadamards give up to 2^k of them. Column weights up to 18 occur for rule 156 at $N = 10$ and up to 243 for rule 150. Where the gate is a reset or an identity the branching is 1 and the column is a single entry.

The diagonal blocks are the strongly connected components , and the strictly upper entries are *drainage*: transitions that leave one SCC for another and can never come back. That asymmetry is the whole content of the triangular form.

A recurrent block is almost never diagonal. This is the expectation worth correcting. "Recurrent" constrains the *set*: a terminal SCC is closed — nothing leaves it — and strongly connected — everything inside reaches everything else. It says nothing whatever about individual states being fixed. A state in an attractor keeps moving; it simply never gets out. Three shapes are possible:

block	shape	diagonal?
fixed point	1×1 with a self-loop	yes — the only diagonal case
pure cycle of length L	$L \times L$ permutation matrix	**no**: zero on the diagonal for $L > 1$
branching attractor	general irreducible sparse block	no

So a diagonal recurrent part means *every attractor is a fixed point*, which is a special case rather than the norm. Counting the terminal blocks of the eight rules drawn here:

rule	tuple	fixed pts	cycles	branching	largest	max col	recurrent part
156	`IIVI`	6	0	138	20	18	not diagonal
157	`EIVI`	1	0	15	8	8	not diagonal
109	`EIIV`	11	0	43	55	55	not diagonal
150	`IVVI`	1	0	5	462	243	not diagonal
158	`IEVI`	1	0	5	10	4	not diagonal
203	`VEII`	22	0	6	2	2	not diagonal
100	`IDIV`	60	0	0	1	1	diagonal
29	`EIVD`	1	0	15	8	8	not diagonal

Two things stand out.

- **Only rule** 100 **has a diagonal recurrent part**, and it is diagonal for the strongest possible reason: *every* strong component of its graph is a singleton ($n_{\mathrm{scc}} = 1024 = \dim$), so the transition graph is acyclic and all 60 attractors are fixed points. Its gold squares are 60 isolated diagonal entries.

- **Not one pure cycle appears in any of the eight.** Every attractor with more than one state is a *branching* block — some column inside it carries several entries — because these are Hadamard rules and a Hadamard puts the state into a superposition of successors that all lie inside the attractor. Rule 203's six two-state attractors are the smallest example: each is a full 2×2 block, not the anti-diagonal 2-cycle one might expect. Rule 156's largest is 20×20 and about two-thirds full.

The empty "cycles" column is a property of the gate, not of recurrence. Replace the Hadamard by X and every rule becomes a permutation: the successor map is single-valued, every column has weight exactly 1, and every attractor *is* a pure cycle — an $L \times L$ permutation block with nothing on its diagonal. That is R10's setting, and it is the one case where the recurrent part of the matrix is as far from diagonal as it can be while still being a bijection.

6 What this does and does not show

- **It is support, not dynamics.** A spy plot cannot distinguish a sector that the dynamics explores ergodically from one it merely fails to leave. The block structure is a statement about which amplitudes can be nonzero, and nothing here measures how the weight moves inside a block.

- **It is one rule and one boundary condition.** Rule 156 was chosen because unitarity makes the partition unambiguous (A2). For a dissipative rule the same picture would need a choice — weak components block-*diagonal*, strong components only block-*triangular* — which is the distinction adjudicated in R-T14 and quantified in R9 §7.3. `pbc` is held pending caveats and is not drawn.
- **The permutation is not canonical.** Blocks are ordered by size and states ascending within a block; any other order inside a sector gives an equally valid block-diagonal picture with a different interior pattern. What is canonical is the *partition*, and therefore the block sizes.
- **It does not prove the asymptotics.** The figure is a finite-N object. That the block count is F_{N+2} is checked to $N = 12$ here and derived in R2; the picture illustrates it and does not establish it.

7 Reproduce

```
python -m qca_fragmentation.viz.spy --rule 156 --N 10 --bc obc0
```

writes `fig_spy_156_N10_obc0`, `fig_spy_156_N8_obc0`, `fig_spy_zoom_156_N10_obc0` (each as `.pdf` and `.png`) and `reports/tex/tab_r12_spy_obc0.tex`. The module is `code/qca_fragmentation/viz/spy.py`; its tests are in `code/qca_fragmentation/tests/test_spy.py`, and they pin the zero off-block count, the Fibonacci block count, the partition agreeing with the Tier-1e `weak_components` pass, and the permutation being a genuine permutation.